

INTRODUCTION TO INFO. & COMM. TECHNOLOGY

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The Hexadecimal System

- Binary code which is understood by computers is hard to read and write
- One way to make reading and writing code easier is to use a shorthand for binary code
- This is why we use the hexadecimal system

The Hexadecimal System

- It uses a base of 16
- The hexadecimal system uses 16 digits
- This system uses uppercase letters to represent the digits 10 through 15
- Therefore, the **hexadecimal digits** are 0 through 9 and A through F

Important to note

- To avoid confusion, numbers in systems other than decimal are read digit by digit
- The number 283 in **decimal** is read “two hundred eighty three”
- The **binary** number 1011 is read “one-zero-one-one”
- The **hexadecimal** number 28A is read “two-eight-A”

Decimal to Hexadecimal

Divide by the base 16 repeatedly

i. $230_{10} = ?$

Decimal to Hexadecimal

Divide by the base 16 repeatedly

i. $230_{10} = ?$

16		230

16		14-----6

		0-----14

		14 = E

Decimal to Hexadecimal

Divide by the base 16 repeatedly

i. $230_{10} = \text{E6}$

ii. $1987_{10} = ?$

Decimal to Hexadecimal

Divide by the base 16 repeatedly

i. $230_{10} = \text{E6}$

ii. $1987_{10} = ?$

16		1987
16		124 ----- 3
		7 ----- 12

$12 = \text{C}$

Decimal to Hexadecimal

Divide by the base 16 repeatedly

i. $230_{10} = \text{E6}$

ii. $1987_{10} = 7\text{C}3$

iii. $15_{10} = ?$

Decimal to Hexadecimal

Divide by the base 16 repeatedly

i. $230_{10} = \text{E6}$

ii. $1987_{10} = 7\text{C3}$

iii. $15_{10} = \text{F}$

iv. $17_{10} = ?$

Decimal to Hexadecimal

Divide by the base 16 repeatedly

i. $230_{10} = \text{E6}$

ii. $1987_{10} = 7\text{C}3$

iii. $15_{10} = \text{F}$

iv. $17_{10} = 11$

v. $1010 = ?$

Decimal to Hexadecimal

Divide by the base 16 repeatedly

i. $230_{10} = \text{E6}$

ii. $1987_{10} = 7\text{C}3$

iii. $15_{10} = \text{F}$

iv. $17_{10} = 11$

v. $1010 = 3\text{F}2$

Decimal to Hexadecimal (using table)

5	4	3	2	1	0
16	16	16	16	16	16
1048576	65536	4096	256	16	1

Converting 89468 into hexadecimal

5	4	3	2	1	0
16	16	16	16	16	16
1048576	65536	4096	256	16	1
	1				

Converting 89468 into hexadecimal

5	4	3	2	1	0
16	16	16	16	16	16
1048576	65536	4096	256	16	1
	1	5			

Converting 89468 into hexadecimal

5	4	3	2	1	0
16	16	16	16	16	16
1048576	65536	4096	256	16	1
	1	5	13		

Converting 89468 into hexadecimal

5	4	3	2	1	0
16	16	16	16	16	16
1048576	65536	4096	256	16	1
	1	5	13	7	

Converting 89468 into hexadecimal

5	4	3	2	1	0
16	16	16	16	16	16
1048576	65536	4096	256	16	1
	1	5	13	7	12

Converting 89468 into hexadecimal

89468

23932

3452

124

12

5

4

3

2

1

0

16

16

16

16

16

16

1048576

65536

4096

256

16

1

1

5

13

7

12

1

5

D

7

C

hexadecimal

Converting 89468 into hexadecimal

89468						
	5	4	3	2	1	0
	16	16	16	16	16	16
23932	1048576	65536	4096	256	16	1
3452		1	5	13	7	12
124		1	5	D	7	C hexadecimal
12						

89468 divided by 65536 = 1.3 --> put 1

89468 - 65536 = 23932

23932 divided by 4096 = 5.8 --> put 5

4096 x 5 = 20480 --> 23932 - 20480 = 3452

3452 divided by 256 = 13.4 --> put 13

256 x 13 = 3328 --> 3452 - 3328 = 124

124 divided by 16 = 7.75 --> put 7

16 x 7 = 112 --> 124 - 112 = 12

12 divided by 1 = 12 --> put 12

Hexadecimal to Decimal

i. $2A8_{16} = ?$

$$2 \times 16^2 + A \times 16^1 + 8 \times 16^0$$

$$2 \times 256 + 10 \times 16 + 8 \times 1$$

$$512 + 160 + 8$$

680 in decimal system

Hexadecimal to Decimal

i. $2A8_{16} = 680$

ii. $1F65_{16} = ?$

Hexadecimal to Decimal

i. $2A8_{16} = 680$

ii. $1F65_{16} = ?$

$$1 \times 16^3 + F \times 16^2 + 6 \times 16^1 + 5 \times 16^0$$

$$1 \times 4096 + 15 \times 256 + 6 \times 16 + 5 \times 1$$

$$4096 + 3840 + 96 + 5$$

8037 in decimal system

Hexadecimal to Decimal

i. $2A8_{16} = 680$

ii. $1F65_{16} = 8037$

Hexadecimal to Decimal

- i. $2A8_{16} = 680$
- ii. $1F65_{16} = 8037$
- iii. $FF_{16} = 255$
- iv. $1097_{16} = 4247$
- v. $66_{16} = 102$

Ref Chapters

- Book 2
- Chp 5